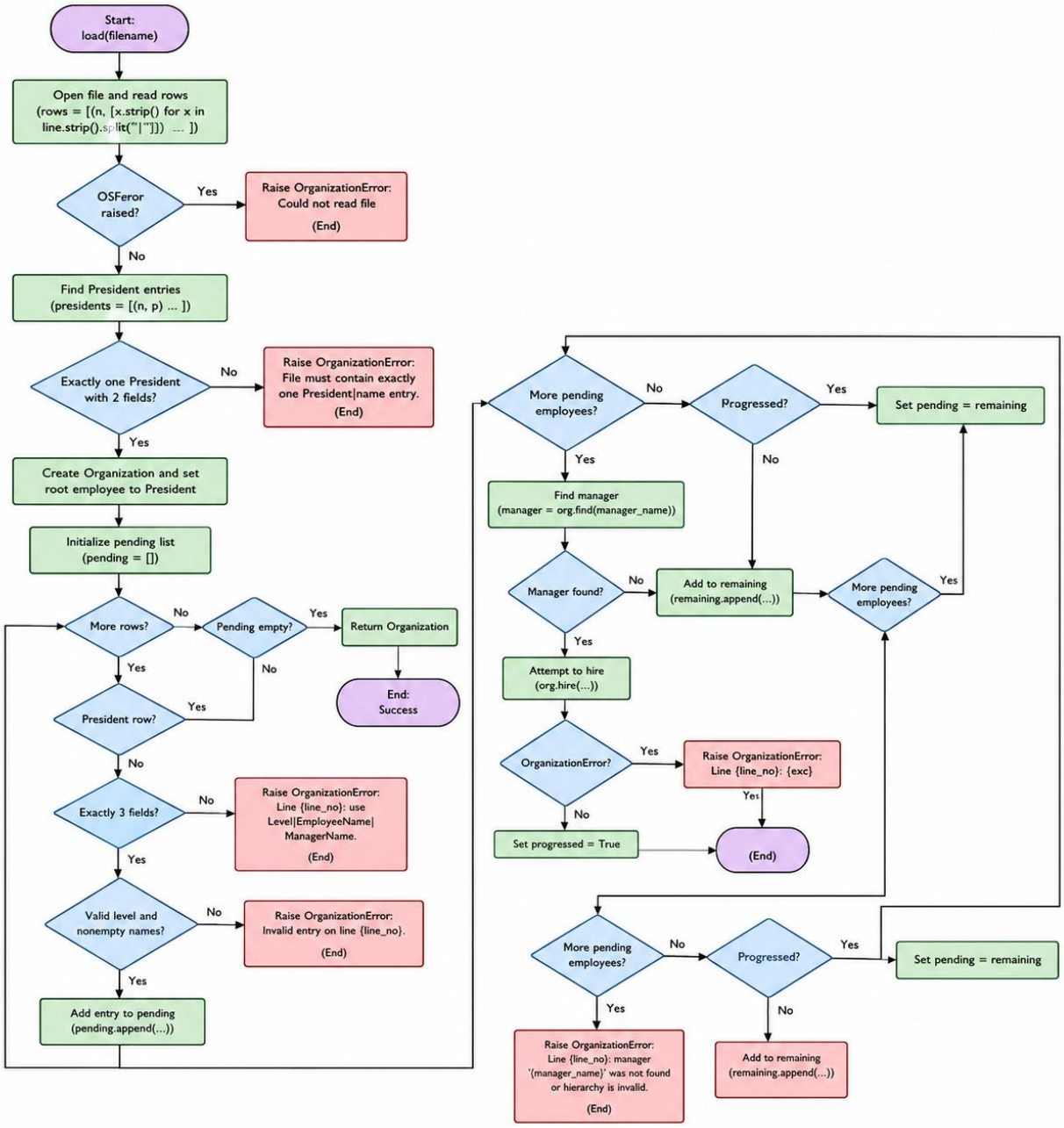

Start: load(filename)
Open file and read rows (rows = [(n, [x.strip() for x in line.strip().split("|")]) ...])
OSFeror raised?
Yes
Raise OrganizationError: Could not read file (End)
No
Find President entries (presidents = [(n, p) ...])
Exactly one President with 2 fields?
No
Raise OrganizationError: File must contain exactly one President|name entry. (End)
Yes
Create Organization and set root employee to President
Initialize pending list (pending = [])
More rows?
No
Pending empty?
Yes
Return Organization
End: Success
No
Yes
President row?
Yes
No
Exactly 3 fields?
No
Raise OrganizationError: Line {line_no}: use Level|EmployeeName|ManagerName. (End)
Yes
Valid level and nonempty names?
No
Raise OrganizationError: Invalid entry on line {line_no}. (End)
Yes
Add entry to pending (pending.append(...))
More pending employees?
No
Progressed?
Yes
Set pending = remaining
Yes
No
Find manager (manager = org.find(manager_name))
Manager found?
No
Add to remaining (remaining.append(...))
More pending employees?
Yes
Yes
Attempt to hire (org.hire(...))
OrganizationError?
Yes
Raise OrganizationError: Line {line_no}: {exc}
Yes
No
Set progressed = True
(End)
More pending employees?
No
Progressed?
Yes
Set pending = remaining
Yes
No
Raise OrganizationError: Line {line_no}: manager '{manager_name}' was not found or hierarchy is invalid. (End)
Add to remaining (remaining.append(...))